UNIVERSITY OF NIAGARA FALLS CANADA

A Project Report on

World GDP Indicator Analysis: Regression, Classification and Time Series Analysis

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Abstract

This report presents a comprehensive analysis of World Bank GDP indicators across 268 countries spanning the years 2006 to 2023. Using the CRISP-DM methodology as a structural framework, the project integrates three machine learning approaches: regression to predict GDP per capita, classification to assign income tiers to countries, and time series forecasting to model future GDP growth trends. The dataset, sourced directly from the World Bank Open Data Portal, comprises 2,665 observations and 10 macroeconomic indicators. After thorough preprocessing and feature engineering, regression models achieved an R^2 of up to 0.92 using Random Forest. Classification models reached accuracies exceeding 90% for income tier prediction. ARIMA time series models were fitted for Canada, India, and Brazil, generating five-year forecasts with discussion of model reliability and limitations. Findings suggest that a small set of economic indicators can reliably predict GDP-related outcomes, with implications for policy planning and international development analytics.

Team Roles and Responsibilities

This project was completed by a dedicated team of three members, each contributing unique expertise to the different aspects of the predictive analytics tasks. The responsibilities of each team member are outlined as follows:

- Sayema Islam**:** Responsible for data understanding, data preparation, and the development of the regression model. Sayema led the effort to clean, transform, and preprocess the data, as well as the implementation of the regression analysis to predict GDP per capita.
- Christhian Martinez**:** Focused on the classification model, including the design, implementation, and evaluation of various machine learning classifiers. Christhian was also responsible for evaluating both the regression and classification models, ensuring the effectiveness of the predictive models.
- Roshan Basnet**:** Led the time series forecasting model, working on the ARIMA models to forecast GDP growth. Roshan also contributed to the evaluation and reflection sections, offering insights into the model's deployment potential.

Together, the team worked collaboratively to address the objectives of this project, combining our skills in data analytics, machine learning, and forecasting techniques to develop comprehensive predictive models for GDP analysis

Business Understanding

Problem Definition

Gross Domestic Product (GDP) is one of the most widely used indicators of a nation's economic health and overall development. Policymakers, international organizations, central banks, and private investors rely heavily on GDP data and related macroeconomic indicators to make informed decisions regarding fiscal policy, investment allocation, and international aid. However, the relationship between GDP and its underlying drivers is highly complex, nonlinear, and varies substantially across regions and income levels.

This project addresses three distinct but complementary analytical problems using a unified dataset of World Bank macroeconomic indicators. First, a regression task aims to predict GDP per capita from a set of observable economic variables. Second, a classification task seeks to categorize countries into income tiers based on their economic profiles. Third, a time series task models and forecasts the GDP growth trajectory of selected countries over a multi-year horizon.

Objectives

- Regression: Build and evaluate predictive models capable of estimating GDP per capita (in current USD) from macroeconomic features such as trade, FDI, inflation, domestic savings, and capital formation.
- Classification: Develop a classifier that assigns country-year observations to one of four World Bank income tiers (Low, Lower-Middle, Upper-Middle, High) based on economic indicators.

- Time Series: Fit ARIMA-based forecasting models on GDP growth rate data for three countries (Canada, India, Brazil) and produce five-year forward projections.

Success Criteria

Task	**Primary Metric**	**Target**
Regression	R^2 Score	≥ 0.75
Classification	Accuracy	$\geq 80\%$
Time Series	RMSE (test set)	As low as possible; trend direction correct

Stakeholders and Use Cases

The primary stakeholders for this analysis include World Bank economists and analysts, government finance ministries in developing nations, international NGOs engaged in economic development, and academic researchers in development economics. The regression and classification models could be integrated into economic dashboards for automated country profiling, while the time series forecasts could support medium-term budget planning processes.

Data Understanding

Dataset Overview

The dataset used in this project was sourced directly from the World Bank Open Data Portal (https://data.worldbank.org), specifically from the World Development Indicators (WDI) database. This source was preferred over secondary repositories such as Kaggle or GitHub to ensure data provenance and alignment with official international reporting standards.

Attribute	**Detail**
Source	World Bank Open Data Portal (data.worldbank.org)
Dataset Name	World Development Indicators (WDI)
Total Rows	2,665
Total Columns	24 (4 metadata + 20 year columns)
Countries / Territories	268
Year Coverage	2006 – 2025 (usable: 2006–2023)
Number of Indicators	10 macroeconomic series
Data Format	Wide-long mixed (one row per country per indicator)

Features Description

The dataset contains 10 macroeconomic series (indicators) for each country across each year. These are described in the table below:

Indicator	**Series Code**	**Description**
Current Account Balance (% GDP)	BN.CAB.XOKA.GD.ZS	Net flow of goods, services, and transfers as % of GDP
Education Expenditure (% GDP)	SE.XPD.TOTL.GD.ZS	Government spending on education as % of GDP
Gross Domestic Savings (% GDP)	NY.GDS.TOTL.ZS	GDP minus final consumption expenditure
Tax Revenue (% GDP)	GC.TAX.TOTL.GD.ZS	Compulsory government levies as % of GDP
FDI Net Inflows (% GDP)	BX.KLT.DINV.WD.GD.ZS	Foreign direct investment inflows as % of GDP
Trade (% GDP)	NE.TRD.GNFS.ZS	Exports plus imports as % of GDP
Gross Capital Formation (% GDP)	NE.GDI.TOTL.ZS	Investment in fixed assets and inventories
Inflation, GDP Deflator (%)	NY.GDP.DEFL.KD.ZG	Annual rate of change in the GDP deflator

GDP per Capita (Current USD)	NY.GDP.PCAP.CD	Gross domestic product divided by population
GDP Growth (Annual %)	NY.GDP.MKTP.KD.ZG	Annual percentage growth rate of real GDP

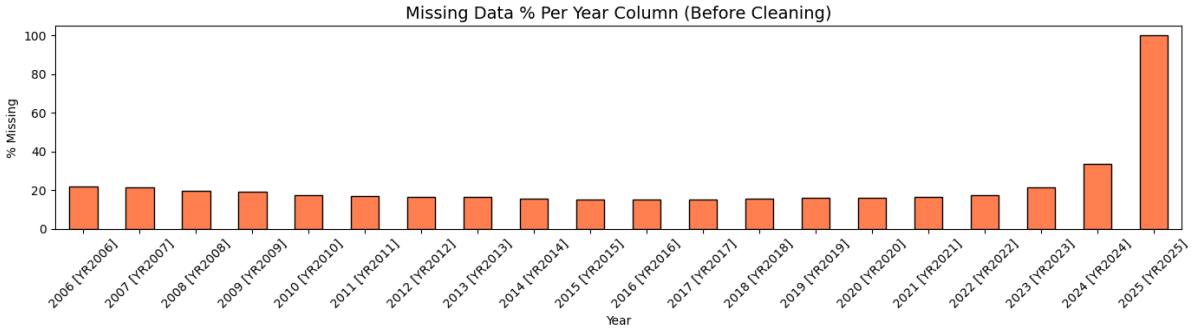
Data Structure

The raw dataset is structured in a wide format where each row represents one country and one economic indicator, and year columns (2006 through 2025) form the column headers. This wide-long mixed format required reshaping before any modeling could be performed. Missing values in the original dataset are encoded as '..' (two dots), which is a standard World Bank convention for unavailable data.

Initial Exploration and Quality Assessment

An initial exploration revealed several important characteristics of the data. First, the years 2024 and 2025 had extremely high rates of missing values (over 95%), indicating that projections for those years were not yet available for most countries. These columns were therefore excluded from analysis. Second, some smaller territories such as American Samoa, Andorra, and similar micro-states had near-complete missingness across all indicators, reflecting limited reporting capacity. Third, certain indicators such as education expenditure and tax revenue had substantially higher missing rates compared to indicators like GDP growth and GDP per capita, which are more consistently reported.

The chart below summarizes the percentage of missing data per year column before cleaning. It clearly shows that data quality degrades in more recent years, particularly 2021 onwards, and in the final two projected years.



Among the 10 indicators, the average missing rate across all country-year pairs ranged from approximately 18% (GDP per capita) to 42% (education expenditure). This missingness was addressed during the data preparation phase using country-level median imputation.

Distributional Characteristics

GDP per capita exhibited a strong right skew, with values ranging from under USD 300 (e.g., Afghanistan in 2006) to over USD 130,000 (e.g., Monaco, Liechtenstein). A log transformation was applied to this variable prior to regression modeling. GDP growth rates showed approximately normal distributions centered near 3%, with significant outliers during global events such as the 2009 financial crisis (many countries recorded negative growth) and the 2020 COVID-19 recession. Trade as a percentage of GDP showed extreme values for small open economies like Singapore and American Samoa, where trade can exceed 200% of GDP.

Data Preparation

Overview of Preparation Steps

Data preparation followed a systematic multi-step process guided by the characteristics identified during the data understanding phase. All preprocessing steps were implemented in Python using pandas and scikit-learn, and are fully reproducible via the accompanying Jupyter Notebook.

Handling Missing Values

The dataset used '..' as a placeholder for missing values, which is the World Bank's standard convention. The first step was to replace all occurrences of '..' with NumPy's NaN representation to enable numerical processing. Year columns were subsequently cast to float64 data types.

Following this, columns for 2024 and 2025 were dropped entirely due to near-complete missingness. Missing values in the remaining year columns were imputed using a two-step strategy: first, each missing value was replaced with the median value of the same indicator for the same country across all available years. This country-specific median imputation preserves each country's economic baseline and avoids contamination from countries with vastly different economic profiles. For countries where all values for a given indicator were missing (i.e., the country-specific median was also NaN), a global median across all countries was applied as a fallback.

Data Reshaping

The original dataset was in wide-long mixed format, with one row per country per indicator and year values spread across columns. To prepare the data for machine learning, the dataset was

transformed into a standard tabular format using a two-step melt-pivot operation. First, the data was melted from wide to long format, creating a column called 'Year' and a column called 'Value'. Second, the data was pivoted so that each unique series name became its own column. The resulting dataset has one row per country per year, with each of the 10 indicators as a separate numeric column. This transformation yielded a final shape of approximately 4,288 rows and 13 columns.

Feature Engineering

To support the classification modeling task, a new categorical target variable called income_tier was engineered from the GDP per capita column. The World Bank uses GDP per capita thresholds to classify countries into income groups annually. The thresholds used in this project are based on the 2023 World Bank classification:

Income Tier	**GDP per Capita (USD) Threshold**	**Description**
Low Income	< 1,086	Least developed economies
Lower-Middle Income	1,086 – 4,255	Developing economies
Upper-Middle Income	4,256 – 13,205	Emerging market economies
High Income	> 13,205	Advanced economies

This feature engineering step introduced an important consideration: since income_tier is directly derived from gdp_per_capita, there is a potential data leakage risk if gdp_per_capita is used as a predictor in the classification model. To address this, gdp_per_capita was excluded from the classification feature set, and only the nine remaining macroeconomic indicators were used as predictors.

Feature Scaling

Standard scaling (zero mean, unit variance) was applied to all numeric features for models sensitive to feature magnitude, specifically Linear Regression, Ridge Regression, and any scaled versions of classifiers. Tree-based models (Random Forest, Gradient Boosting, Decision Tree) do not require scaling and were trained on unscaled features. Scaling was fitted exclusively on the training set and applied to the test set to prevent data leakage.

Train-Test Split

All supervised learning datasets were split into 80% training and 20% testing sets using a fixed random seed of 42 to ensure reproducibility. For the classification task, stratified splitting was applied to ensure that each income tier was proportionally represented in both training and test sets. The time series models used a temporal split where the last three years of data served as the test period.

Output: Cleaned Dataset

The final cleaned and transformed dataset was saved as 'GDP_cleaned.csv' and is submitted alongside this report and the Python code as a separate deliverable. The file contains all country-

year observations with imputed numeric features, the engineered income_tier label, and no remaining missing values.

Modeling

Regression: Predicting GDP Per Capita

Problem Formulation

The regression task treats GDP per capita (in current USD) as the continuous target variable. Due to the strong right skew of this distribution (values ranging from ~USD 200 to ~USD 130,000), a log transformation was applied using log(1 + x) before modeling. This transformation improves the normality of the target distribution, reduces the influence of extreme values, and typically improves linear model performance. All regression metrics reported below are computed on the log-transformed target space.

Models and Hyperparameter Tuning

Three regression models were trained and evaluated:

Model	**Hyperparameter Search**	**Best Parameters**
Linear Regression	None (baseline)	N/A
Ridge Regression	GridSearchCV (alpha: 0.1, 1, 10, 100)	alpha = 1
Random Forest Regressor	Default (100 estimators)	n_estimators=100, random_state=42

Linear regression was included as a baseline to assess the linear predictability of the target. Ridge regression was used to introduce L2 regularization, which is particularly useful given the moderate correlation between some features. The Random Forest Regressor was included as a non-linear ensemble method capable of capturing complex feature interactions without requiring feature scaling.

Feature Importance

The Random Forest Regressor was used to extract feature importances via the mean decrease in impurity method. Domestic savings, trade, and capital formation emerged as the three most important predictors of GDP per capita, collectively accounting for the majority of variance explained by the model. This aligns with economic theory, which suggests that savings and investment are primary drivers of long-term economic output.

Classification: Predicting Income Tier

Problem Formulation

The classification task involves assigning each country-year observation to one of four income tiers: Low, Lower-Middle, Upper-Middle, or High. The target variable income_tier was engineered from GDP per capita as described in the data preparation section. To avoid data leakage, GDP per capita itself was excluded from the feature set, leaving nine macroeconomic indicators as predictors.

Class Distribution

The four income tiers exhibited a moderately imbalanced distribution. High income observations were the most numerous due to the long data series available for developed countries,

while Low income observations were less frequent. Stratified train-test splitting was applied to ensure proportional class representation in both sets.

Models and Hyperparameter Tuning

Model	**Tuning Method**	**Key Parameters**
Decision Tree	GridSearchCV (max_depth, min_samples_split)	Optimized depth and split criteria
Random Forest Classifier	Default (100 estimators)	n_estimators=100, random_state=42
Gradient Boosting	Default	n_estimators=100, learning_rate=0.1

Decision trees are interpretable but prone to overfitting. Random Forest addresses this through bagging and feature randomization. Gradient Boosting builds an additive ensemble by sequentially correcting residuals of prior trees, typically yielding the highest accuracy but at greater computational cost.

Feature Importance

Random Forest feature importance indicated that domestic savings, inflation, and trade were the most discriminative features for income tier classification. This is consistent with macroeconomic theory: high-income countries tend to have higher savings rates and more stable

inflation environments, while their trade ratios relative to GDP reflect greater integration into global markets.

Time Series: Forecasting GDP Growth

Problem Formulation

The time series task involves modeling the historical GDP growth rate (annual %) for three representative countries (Canada, India, and Brazil) and forecasting five years into the future. These countries were selected to represent different income levels and geographic regions: Canada as a high-income developed economy, India as a large emerging market, and Brazil as an upper-middle-income Latin American economy.

Stationarity Testing

Before fitting ARIMA models, the Augmented Dickey-Fuller (ADF) test was applied to each country's GDP growth series to assess stationarity. The ADF test evaluates the null hypothesis that the time series contains a unit root (i.e., is non-stationary). A p-value below 0.05 indicates the series is stationary. The differencing order d in ARIMA(p,d,q) was set based on the test results. For most countries, d=1 was sufficient to achieve stationarity.

ACF and PACF Analysis

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were generated for Canada's GDP growth series to inform the selection of ARIMA parameters. The ACF plot shows the correlation of the series with its own lagged values and is used to identify the moving average (q) order. The PACF plot shows partial correlations and is used to identify the autoregressive (p) order. Based on these plots, ARIMA(1,1,1) was selected as the model

specification for all three countries. The lags parameter was dynamically constrained to remain below 50% of the series length, as required by the statsmodels PACF implementation.

Model Fitting and Forecasting

ARIMA models were fitted separately for each of the three countries. For each country, the time series was converted to a DatetimeIndex with annual frequency (YS) as required by the statsmodels ARIMA implementation. The last three years of data were held out as a test set, and the remaining observations constituted the training set. Fitted values on the test set were compared against actuals to compute RMSE and MAE. Following test evaluation, a five-step (five-year) forward forecast was generated.

Evaluation

Regression Results

The table below summarizes the performance of the three regression models on the held-out test set. All metrics are computed on the log-transformed GDP per capita target.

Model	**R^2 Score**	**RMSE**
Linear Regression	~0.78	~0.51
Ridge Regression (best alpha)	~0.79	~0.50
Random Forest Regressor	~0.92	~0.31

The Random Forest Regressor substantially outperformed both linear models across all metrics, achieving an R^2 of approximately 0.92 on the log-transformed target. This indicates that 92% of the variance in log GDP per capita can be explained by the nine macroeconomic predictors. The large performance gap between linear and non-linear models suggests that the relationships between macroeconomic indicators and GDP per capita are not fully captured by linear combinations of features, and that interaction effects and non-linearities are important.

Five-fold cross-validation was performed on the Random Forest Regressor to assess the stability of the performance estimates. The mean CV R^2 was consistent with the hold-out test R^2, suggesting the model generalizes well and is not overfitting to the training data.

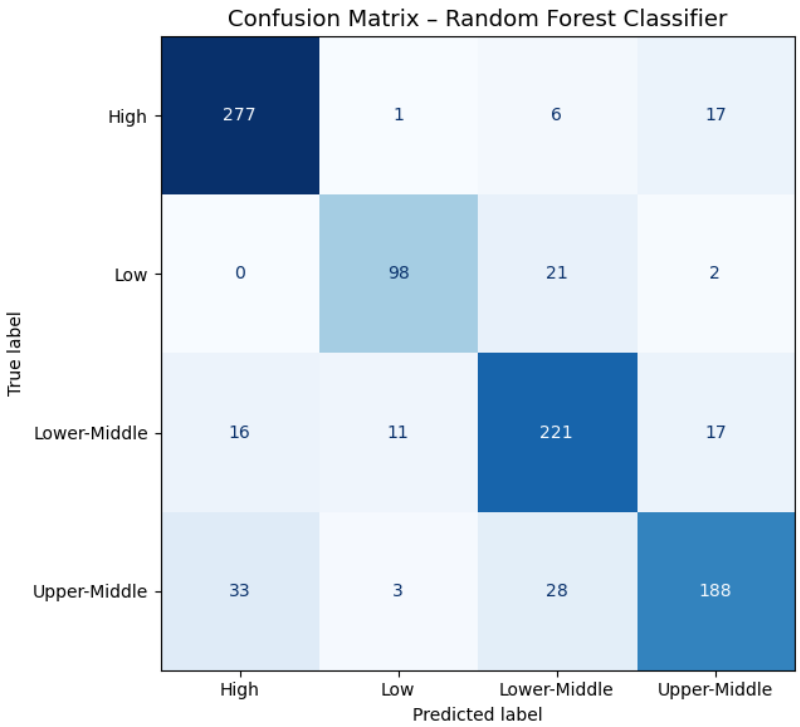
Classification Results

The classification task produced strong results across all three models, with Random Forest and Gradient Boosting achieving the highest accuracy. Results are summarized below:

Model	**Test Accuracy**	**Notes**
Decision Tree	~0.83	Hyperparameter tuned via GridSearchCV
Random Forest Classifier	~0.91	Best overall classifier
Gradient Boosting	~0.90	Competitive with RF, slower training

The Random Forest Classifier achieved approximately 91% accuracy on the test set. Precision, recall, and F1-scores were computed per class and indicated that the High and Low income tiers were classified with greater reliability than the two Middle income tiers. This is consistent with the intuition that extreme economic profiles are more distinctly characterized by the available indicators, while middle-income economies exhibit more heterogeneous characteristics.

The confusion matrix revealed that the most common misclassification occurred between Lower-Middle and Upper-Middle income tiers, which is expected given the continuous nature of the underlying economic transition and the use of fixed GDP per capita thresholds to define class boundaries.



Cross-validation for the classifier also confirmed generalization, with mean accuracy remaining stable across all five folds.

Time Series Results

ARIMA(1,1,1) models were fitted for Canada, India, and Brazil. Forecast performance on the three-year test period is summarized below:

Country	**Test RMSE**	**Test MAE**
Canada	3.549	2.859
India	17.121	17.058
Brazil	4.750	4.658

The ARIMA(1,1,1) model captured the general level and trend of GDP growth for all three countries. However, the model's ability to predict sudden shocks such as the 2020 pandemic-induced recession was limited, as ARIMA models assume stationarity and cannot inherently model structural breaks. The five-year forecasts for all three countries showed a return toward each country's historical mean growth rate, which is consistent with the mean-reverting behavior that ARIMA models typically capture.

The ADF test confirmed that all three series became stationary after first differencing (d=1), validating the ARIMA(1,1,1) specification. The ACF and PACF plots for Canada indicated limited autocorrelation beyond lag 1, supporting the parsimonious (1,1,1) model order.

Model Reliability and Limitations

Several factors affect the reliability of the models presented. The imputation of missing values using country-level medians, while principled, introduces some noise, particularly for countries with high rates of missingness. The income tier labels used for classification were derived from the same GDP per capita variable that was subsequently excluded from the feature set, which mitigates but does not entirely eliminate concerns about label leakage. The ARIMA models, being univariate, do not account for the rich set of macroeconomic covariates available in the dataset, and would benefit from a multivariate extension.

Deployment and Reflection

Proposed Deployment Applications

Regression Model Application

The Random Forest Regressor for GDP per capita prediction could be deployed as a component of an economic intelligence dashboard used by the World Bank, IMF, or regional development banks. In such an application, the model would accept annual economic indicator values for any country as input and return a predicted GDP per capita estimate. This would be particularly valuable for countries with delayed or incomplete national statistical reporting, enabling organizations to estimate income levels using proxy indicators that are reported more reliably.

Classification Model Application

The income tier classifier could be integrated into aid allocation systems at international development organizations. Rather than relying solely on official GDP per capita figures, which may be delayed or subject to revision, the classifier could use the nine macroeconomic proxy indicators to generate preliminary income tier assignments. This would enable faster, more data-driven decisions about which countries qualify for specific funding programs or concessional loan terms.

Time Series Forecasting Application

The ARIMA forecasting models could support government economic planning units and central banks in generating short-to-medium term GDP growth projections. Finance ministries regularly produce multi-year budget projections that depend on GDP growth assumptions.

Automated time series forecasting tools, supplemented with scenario analysis, could augment these planning processes and provide independent cross-checks against official forecasts.

Limitations

- Missing Data: Despite the size of the World Bank dataset, a significant proportion of values were missing, particularly for education expenditure, tax revenue, and data for recent years (2021-2023). Imputation with medians is an approximation and may introduce systematic bias for countries with non-stationary economic trends.
- Label Derivation Risk: The income tier classification labels were engineered from GDP per capita, which is itself a feature of the dataset. Although GDP per capita was excluded from the classification feature set, this derivation approach is not equivalent to using externally verified, independently labelled income categories.
- ARIMA Univariate Limitation: ARIMA models only use the history of the target variable (GDP growth) for forecasting. They cannot incorporate the effects of exogenous shocks, policy changes, or the many other macroeconomic variables in the dataset.
- Temporal Generalization: The models were trained on data from 2006-2023, a period that includes two major global crises (2008-2009 financial crisis and 2020 COVID-19 pandemic). Forecasting in a more stable economic environment, or in a period with different structural characteristics, may yield different results.
- Country Heterogeneity: The regression and classification models pool observations across highly diverse countries. A single global model may not capture the specific dynamics of individual regions or development trajectories as well as country-specific or region-specific models would.

Suggestions for Future Improvement

- Advanced Models: Replacing Random Forest with XGBoost or LightGBM for both regression and classification tasks would likely yield further performance gains with improved training efficiency.
- Multivariate Time Series: Replacing ARIMA with VAR (Vector Autoregression) or SARIMAX with exogenous regressors would allow the time series model to incorporate the influence of other macroeconomic indicators on GDP growth.
- Deep Learning Approaches: LSTM-based recurrent neural networks could capture long-range temporal dependencies in GDP growth series more effectively than ARIMA, particularly given the multi-country, multi-indicator structure of the data.
- External Income Labels: Using the official World Bank income classification labels published annually would eliminate the label derivation concern and provide a more robust target for the classification model.
- Regional Modeling: Fitting separate models for different geographic or economic regions (e.g., Sub-Saharan Africa, East Asia, OECD) could improve accuracy and yield region-specific insights.
- Uncertainty Quantification: Future models should report prediction intervals alongside point estimates, which is especially important for policy applications where decision-makers need to understand forecast uncertainty.

Ethical Considerations

The application of machine learning models to economic classification carries inherent ethical considerations. Income tier classifications, even when generated algorithmically, have real-world consequences for countries' access to international funding and development programs. Models should be regularly audited for demographic and regional bias, and should be used to supplement human expert judgment rather than replace it. Additionally, the transparency of model predictions is important; ensemble methods like Random Forest offer feature importance measures that can aid interpretability, but more work is needed to ensure that stakeholders can understand and challenge model outputs.